

Technical Review Handbook

Portfolio V2.2

Metrics, leakage controls, horizon-aware features, walk-forward validation, Hybrid weighting, uncertainty, and governance

Document	Scope
Purpose	Portfolio review, learning, and technical handoff
Data	Synthetic branch IDs B01-B05 only
Models	ARIMA, ETS, recursive XGBoost, adaptive Hybrid
Version	Portfolio V2.2 - July 2026

From a model that runs to a forecast that can be trusted.

1 System summary

Portfolio V2.2 forecasts monthly warranty revenue by branch for horizons one through six. The technical emphasis is not algorithm novelty alone: it is time-valid evaluation, explicit failure conditions, and traceable model decisions.

Trust equation

Trustworthy forecast = valid data x temporal validation x common samples x stable evidence x auditable fallback.

Component roles

Model	Role	Main risk
ARIMA	Autocorrelation, differencing, linear time patterns	Limited nonlinear interactions
ETS	Level, trend, and seasonal smoothing	May lag structural change
XGBoost	Nonlinear lag and calendar interactions	Leakage and recursive error
Hybrid	Diversifies model-specific misses	Optimistic weights if evaluation is not nested

Pipeline

Stage	Control
Input	Schema, duplicate-key, missing-value, and open-month checks
Features	Lagged history only; blocked forecast-time fields
Backtest	Expanding origins and $h=1,\dots,6$
Comparison	Exactly three component models on common keys
Hybrid	Weights from earlier origins only
Interval	Direct historical Hybrid residual scale
Output	Route, weight source, interval source, and sample counts

2 Forecast metrics

Metric	Formula	Question answered	Caution
RMSE	$\sqrt{\text{mean}((\text{Pred} - \text{Actual})^2)}$	How strongly should large misses be penalized?	Sensitive to outliers
MAE	$\text{mean}(\text{abs}(\text{Pred} - \text{Actual}))$	What is a typical absolute miss?	Does not emphasize spikes
MAPE	$\text{mean}(\text{abs}(\text{error} / \text{Actual})) \times 100$	How large is error relative to Actual?	Unstable near zero
WAPE	$\text{sum}(\text{abs}(\text{error})) / \text{sum}(\text{abs}(\text{Actual})) \times 100$	What is aggregate relative error?	Large months dominate
Bias	$\text{mean}(\text{Pred} - \text{Actual})$	Is the system systematically high or low?	Positive and negative errors cancel
n	Valid prediction count	How much evidence supports the score?	Small n creates unstable rankings

Interpretation sequence

- First verify no leakage and identical evaluation keys.
- Use RMSE as the primary ranking metric.
- Use MAE to understand the typical month.
- Use MAPE and WAPE to add relative-scale context.
- Inspect Bias, sample size, branch, horizon, and worst months.
- Confirm improvements with an outer test or bootstrap evidence.

Important

A lower RMSE does not prove the model is deployable. It may still be caused by target leakage, easier samples, repeated use of the same test period, or an unavailable feature.

3 Leakage and feature availability

Leakage type	Example	V2.2 control
Target	Same-month YoY growth algebraically contains revenue	Explicit blocked-feature list
Temporal	Rolling calculation sees future test rows	Lag before rolling; fold-local training
Availability	Future claim count used at h=3	Horizon-aware routes
Selection	Full-history importance chooses features before test	Nested feature experiments

Public availability matrix

Feature family	h=1	h=2-4	h=5-6
Calendar	Known	Known	Known
Revenue lags	Recursive	Recursive	Recursive
Rolling revenue	Recursive	Recursive	Recursive
Lag-12 revenue	Known	Known	Known
Same-month growth	Blocked	Blocked	Blocked
Future activity counts	Blocked	Blocked	Blocked

Horizon routes

- recursive_core: compact lag, rolling, trend, growth, volatility, and calendar features.
- safe_compact: core plus lag-6 and lag-12 signals for longer horizons.
- guardrail route: conservative core set when experiments do not justify switching.
- short-history route: explicit lower evidence threshold with visible fallback metadata.

Open month rule

The current incomplete month is removed before feature engineering. Otherwise a partial target contaminates lags, rolling windows, and the backtest.

4 Time-series validation

Expanding walk-forward

At each origin, the model trains only on earlier months and predicts $h=1, \dots, 6$. The origin then advances. Random row splitting is not used because it would mix past and future.

Common-key contract

- Every branch-origin-date-h key has ARIMA, ETS, and XGB.
- Actual is unique within the key.
- Predictions and weights are not missing.
- No model appears twice.
- The pipeline stops instead of silently dropping incomplete models.

Independent Hybrid evaluation

For an evaluation origin t , Hybrid weights are estimated from origins earlier than t . Evidence priority is prior branch-horizon errors, then prior branch errors, then equal-weight cold start.

Evidence level	When used	Output label
Branch + horizon	Enough prior observations for all three models	prior_branch_h
Branch	Horizon evidence is sparse	prior_branch
Cold start	No prior residuals exist	equal_weight_cold_start

Why this matters

Estimating weights and scoring them on the same residuals makes Hybrid performance optimistic. Prior-origin weighting separates the decision from the observation being evaluated.

5 XGBoost configuration

Parameter	Reference	Meaning
objective	reg:squarederror	Squared-error regression
eval_metric	rmse	Monitoring metric
eta	0.05	Learning rate per boosting round
max_depth	4	Maximum tree interaction depth
min_child_weight	3	Minimum evidence for a child node
subsample	0.8	Row sampling per tree
colsample_bytree	0.8	Feature sampling per tree
nrounds	200 production reference	Maximum boosting rounds
seed	20260717	Reproducible stochastic sampling

Feature importance

Field	Meaning	Caution
Gain	Average loss reduction from the feature	Not causality or deployment permission
Cover	Observations affected by splits	Reach is not predictive gain
Frequency	How often the feature is used	Frequent use can still be weak

Parameter governance

- Tune only after the final leakage-safe feature set is chosen.
- Use nested walk-forward selection rather than random cross-validation.
- Compare fixed baseline, selected features, and tuned features separately.
- Do not promote a parameter set solely because one holdout month improves.

6 Hybrid and uncertainty

Inverse-RMSE weighting

$\text{weight_m} = (1 / \text{RMSE_m}) / \text{sum_j}(1 / \text{RMSE_j})$

$\text{Hybrid prediction} = \text{sum_m}(\text{weight_m} \times \text{prediction_m})$

Weights are normalized and must sum to one. Missing weights are an error, not a value to be ignored with na.rm.

Residual-based interval

$\text{lower95} = \max(0, \text{prediction} - 1.96 \times \text{Hybrid RMSE})$

$\text{upper95} = \text{prediction} + 1.96 \times \text{Hybrid RMSE}$

Hybrid RMSE is calculated from reconstructed historical Hybrid residuals. This preserves correlation among ARIMA, ETS, and XGB errors better than summing independent component variances.

Interval source	Use
branch_h_residual	At least three Hybrid residuals at the same branch and horizon
branch_residual_fallback	Horizon evidence is sparse; use the branch-level scale

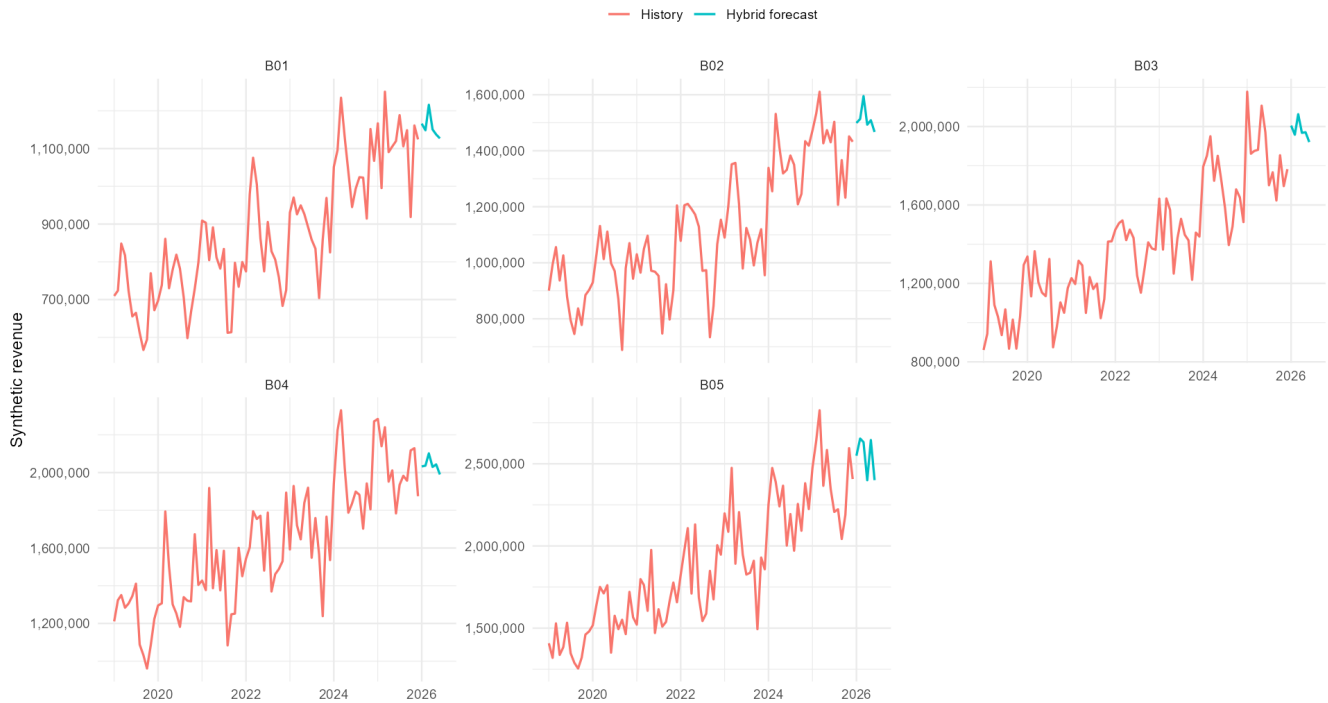
Interval limitation

A $1.96 \times \text{RMSE}$ interval is an approximation. Production monitoring should measure empirical coverage and width by branch and horizon.

7 Synthetic demonstration

The repository includes a reproducible generator for five synthetic branches. The chart below is created by the same pipeline used in the automated smoke test. It contains no company data and is not a performance claim.

Synthetic warranty revenue: history and Hybrid forecast
Portfolio demonstration only - no company data



Reproducibility

Run `scripts/run_demo.R` from the repository root. The demo generates input, backtests all models, builds prior-origin Hybrid weights, forecasts six months, and writes Excel and PNG outputs.

8 Governance checklist

Layer	Required evidence
Data	Completed months, unique keys, required fields, no missing target
Availability	Every feature known at its forecast horizon
Backtest	Expanding time order and common model keys
Selection	Feature and parameter decisions use earlier data only
Performance	RMSE plus MAE, MAPE, WAPE, Bias, and n
Stability	Branch, horizon, worst-month, and resampling checks
Hybrid	Three models, weights sum to one, visible fallback source
Uncertainty	Residual source and empirical coverage monitoring
Operations	Version, seed, configuration, tests, and rollback

Champion, Challenger, Fallback

- Champion: the current evidence-backed production choice.
- Challenger: promising, but still under independent evaluation.
- Fallback: conservative behavior when history or model components are incomplete.

Remaining research

- Empirical interval calibration.
- Feature and residual drift monitoring.
- Nested feature and hyperparameter selection.
- Operational pipeline nowcasting with genuinely available current-period signals.
- Formal promotion and rollback thresholds.

Final principle

A model is not permanently good because it once won. Preserve versions, rerun backtests, monitor drift, and allow a simpler model to become Champion again.

Source: Warranty Revenue Forecasting System Portfolio V2.2. All examples are synthetic and anonymized.